

TOPIC

OBSTACLE AVOIDANCE USING ARTIFICIAL POTENTIAL FIELD



SUBJECT
ROBOTICS

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DETECT
OBSTACLES



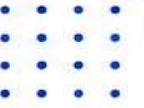
CALCULATE
FORCES



MOVE
SAFELY



REACH
GOAL



What is Obstacle Avoidance in Robotics?

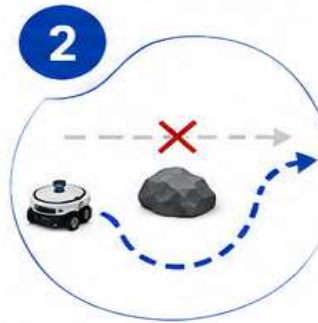
Obstacle avoidance enables a robot to move toward its target while **preventing collisions** with objects in its environment.



1

Robot Perception

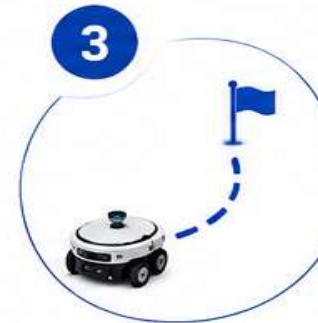
The robot uses sensors (like LiDAR, ultrasound, or camera) to detect obstacles in its surroundings.



2

Path Adjustment

When an obstacle is detected, the robot adjusts its direction or path to avoid collision.



3

Safe Navigation

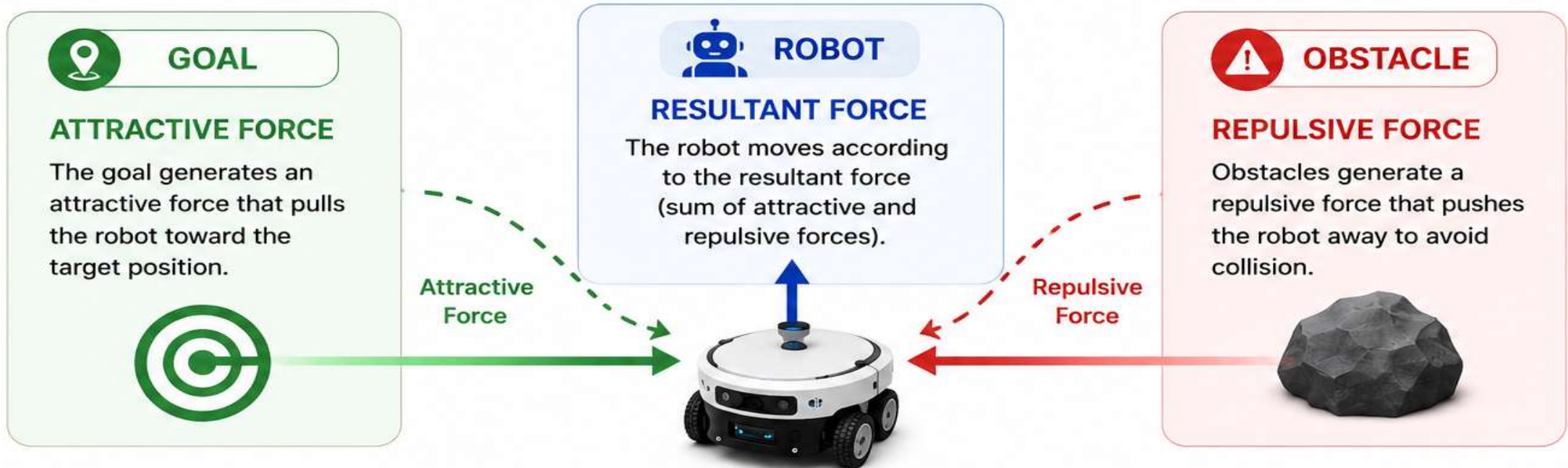
The robot continues toward the target safely by avoiding obstacles in its environment.



The main objective of obstacle avoidance is to ensure **safe, efficient,** and **reliable navigation** of the robot in dynamic environments.

What is the Artificial Potential Field Method?

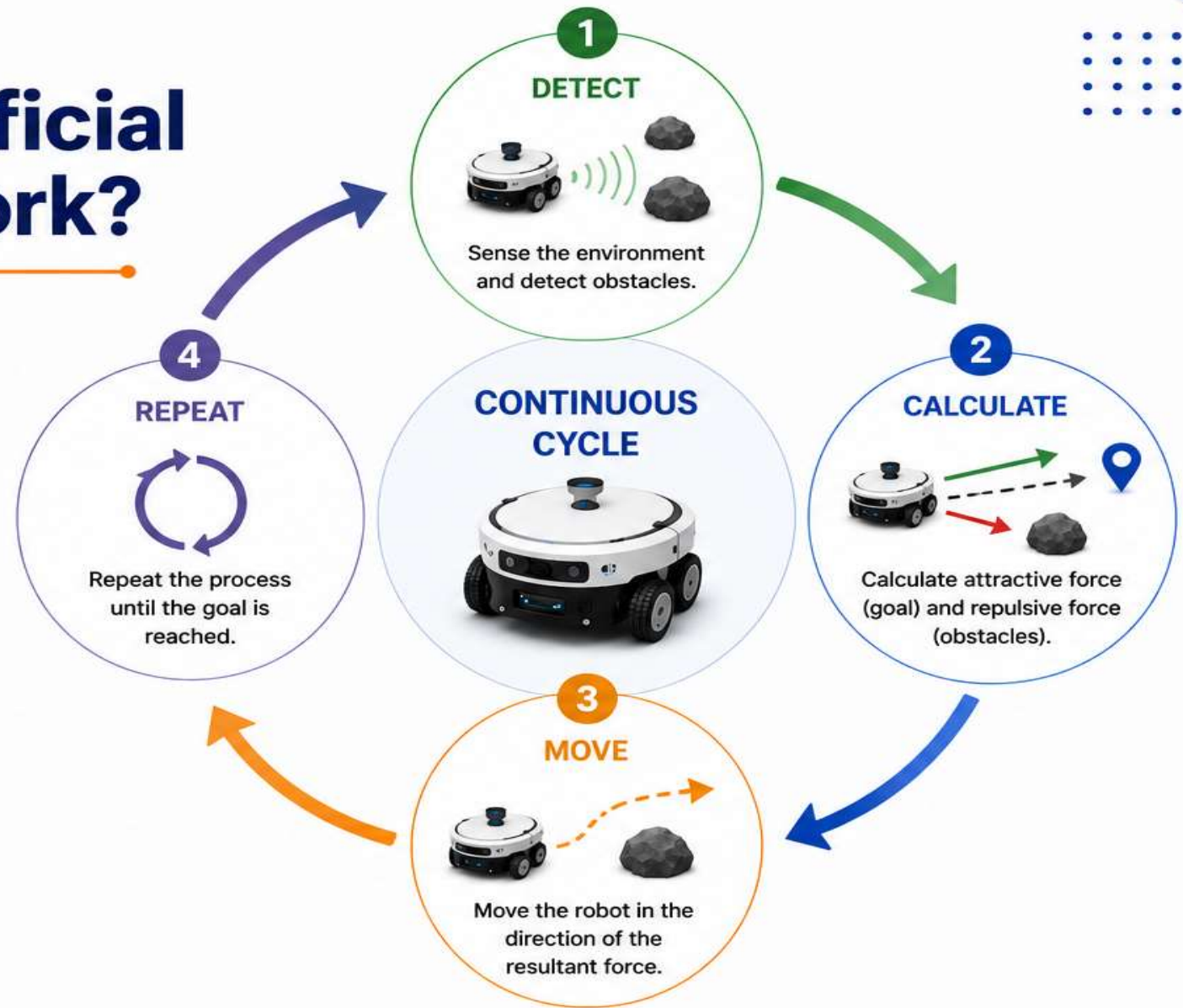
Artificial Potential Field (APF) is a motion-planning technique in which the robot is guided by **artificial forces** generated by the **goal** and **obstacles**.



The robot is **attracted** toward the goal and **repelled** by obstacles. The combination of these forces provides a **safe path to the goal**.

How Does an Artificial Potential Field Work?

APF works in a **continuous cycle**. At every step, the robot senses the environment, calculates the forces, moves in the required direction, and repeats the process until the goal is reached.



1

DETECT

The robot senses the surrounding environment using its sensors.



2

CALCULATE

The robot calculates attractive and repulsive forces acting on it.



3

MOVE

The robot moves in the direction of the resultant force.



4

REPEAT

The process repeats continuously until the robot reaches the goal.



The robot continuously updates its decision based on the current situation, making it a **reactive and real-time navigation** method.

How Does the Goal Attract the Robot?

Attractive Potential Field

The goal generates an attractive potential field that **pulls** the robot toward the target position.



MATHEMATICAL MODEL

- Attractive Potential Function

$$U_{att}(\mathbf{q}) = \frac{1}{2} k_{att} \|\mathbf{q} - \mathbf{q}_g\|^2$$

Where,

- $\mathbf{q}$ = current position of the robot
- $\mathbf{q}_g$ = goal position
- k_{att} = attractive gain (positive constant)
- $\|\mathbf{q} - \mathbf{q}_g\|$ = distance between robot and goal

- Attractive Force

$$\mathbf{F}_{att}(\mathbf{q}) = -\nabla U_{att}(\mathbf{q}) = -k_{att}(\mathbf{q} - \mathbf{q}_g)$$



The farther the robot is from the goal, the **stronger** the attractive force.

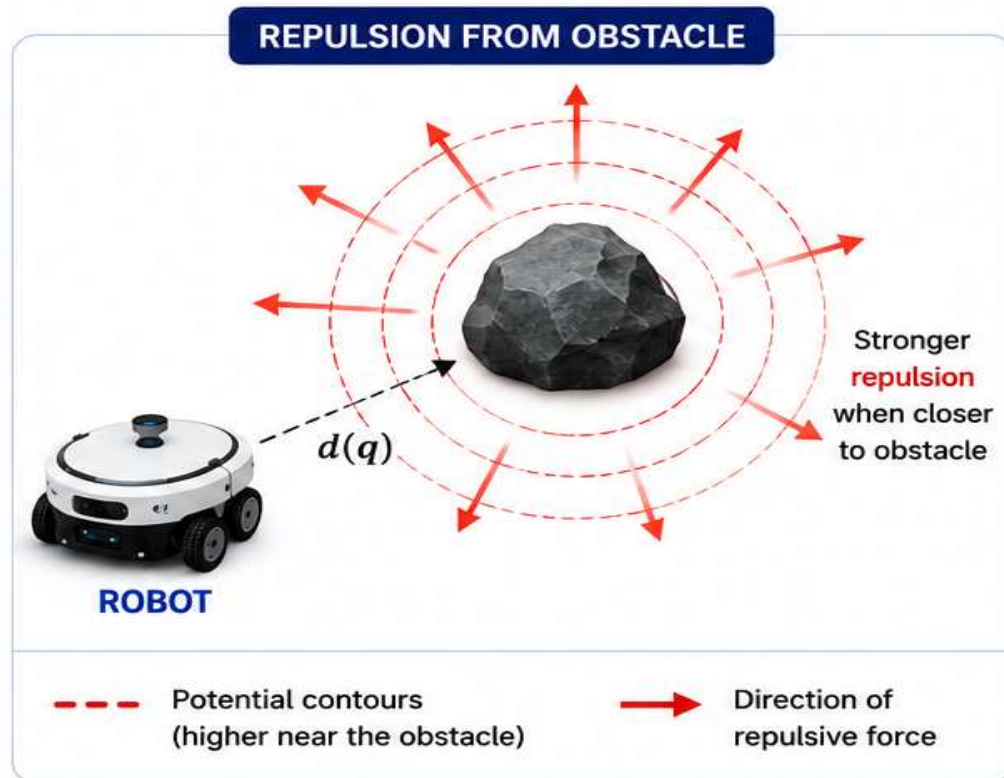


The attractive potential always **pulls** the robot toward the goal, helping it to **reach the target position**.

How Does the Robot Avoid Obstacles?

Repulsive Potential Field

Obstacles generate a repulsive potential field that **pushes** the robot **away** when it comes sufficiently close.



MATHEMATICAL MODEL

• Repulsive Potential Function

$$U_{rep}(q) = \begin{cases} \frac{1}{2} k_{rep} \left(\frac{1}{d(q)} - \frac{1}{d_0} \right)^2, & d(q) \leq d_0 \\ 0, & d(q) > d_0 \end{cases}$$

Where,

$d(q)$ = distance between robot and nearest obstacle

d_0 = influence distance of the obstacle

k_{rep} = repulsive gain (positive constant)

$U_{rep}(q)$ = repulsive potential

• Repulsive Force

$$F_{rep}(q) = -\nabla U_{rep}(q)$$



As the robot gets closer to the obstacle (d decreases), the repulsive force **increases** rapidly.

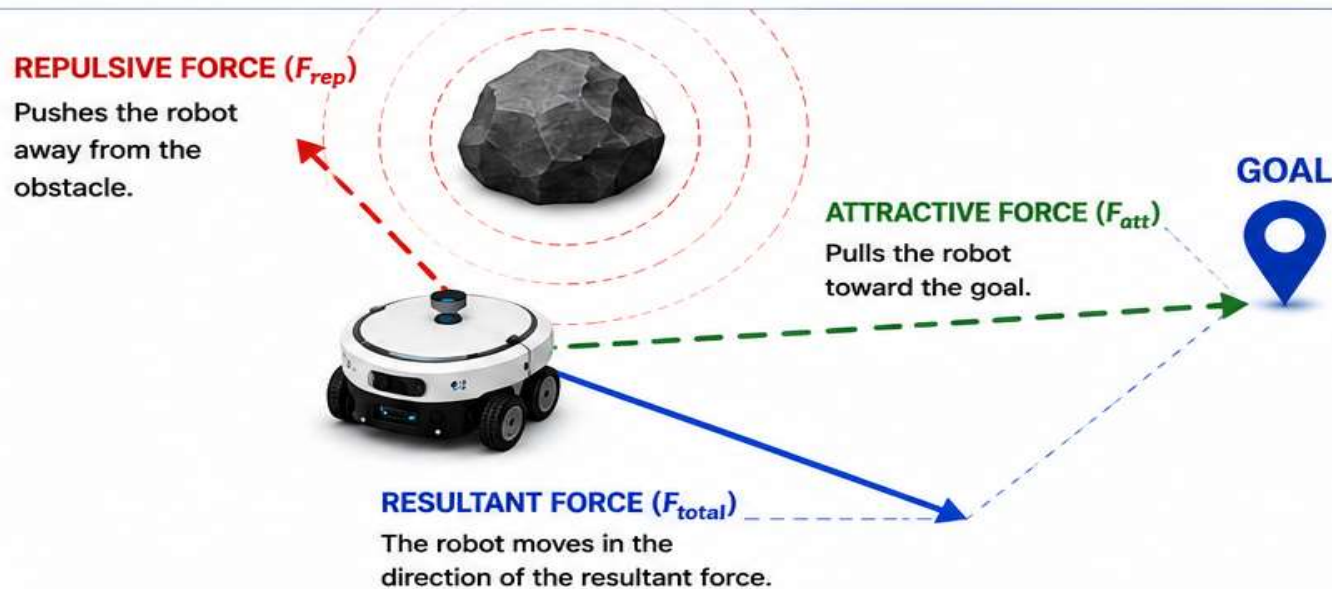


The repulsive potential **prevents** the robot from collisions by creating a **strong force away** from obstacles when the robot is within the influence distance (d_0).

How Does the Robot Choose Its Direction?

Resultant Force Determines the Robot's Motion

The robot is influenced by both **attractive force** (toward the goal) and **repulsive force** (away from obstacles). The combination of these forces gives the **resultant force**, which decides the direction of motion.



The robot does not move directly toward the goal or away from the obstacle. It moves in the direction of the **resultant force**.

MATHEMATICAL MODEL

Total Potential

$$U(q) = U_{att}(q) + U_{rep}(q)$$

Total Force

$$F_{total} = F_{att} + F_{rep}$$

Motion Update

$$\dot{q} = k F_{total}$$

where k = positive gain
 $\dot{q}$ = robot velocity



Key Point

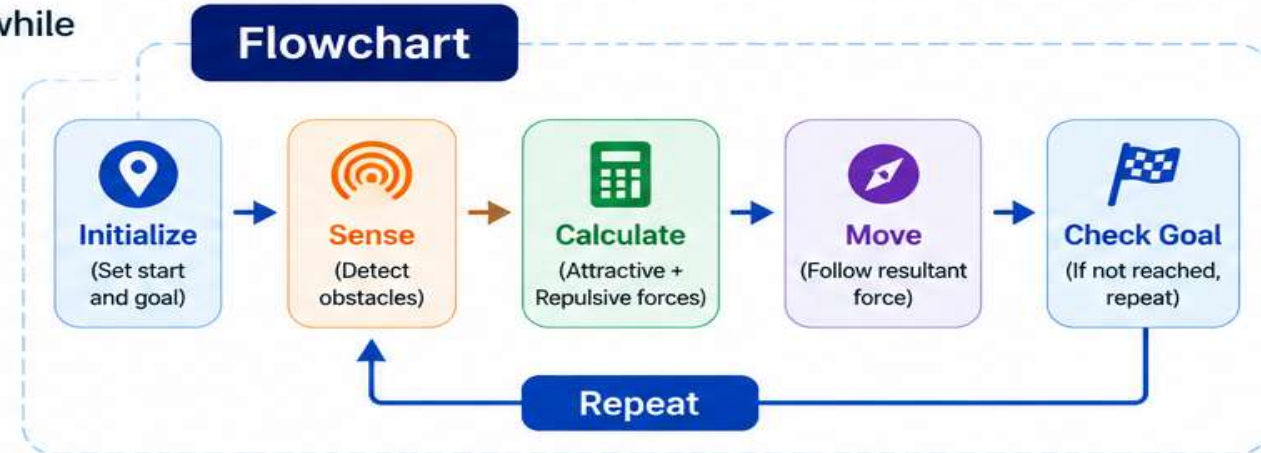
- **Attractive force** guides the robot toward the goal.
- **Repulsive force** keeps the robot away from obstacles.
- The **resultant force** decides the actual direction.
- Robot always follows the **resultant force**.



The direction of the robot is determined by the **resultant force**, which is the **combination** of **attractive** and **repulsive** forces acting in the environment.

What is the APF Navigation Algorithm?

The APF algorithm continuously **senses** the environment, **calculates** the forces and **moves** the robot towards the goal while avoiding obstacles.

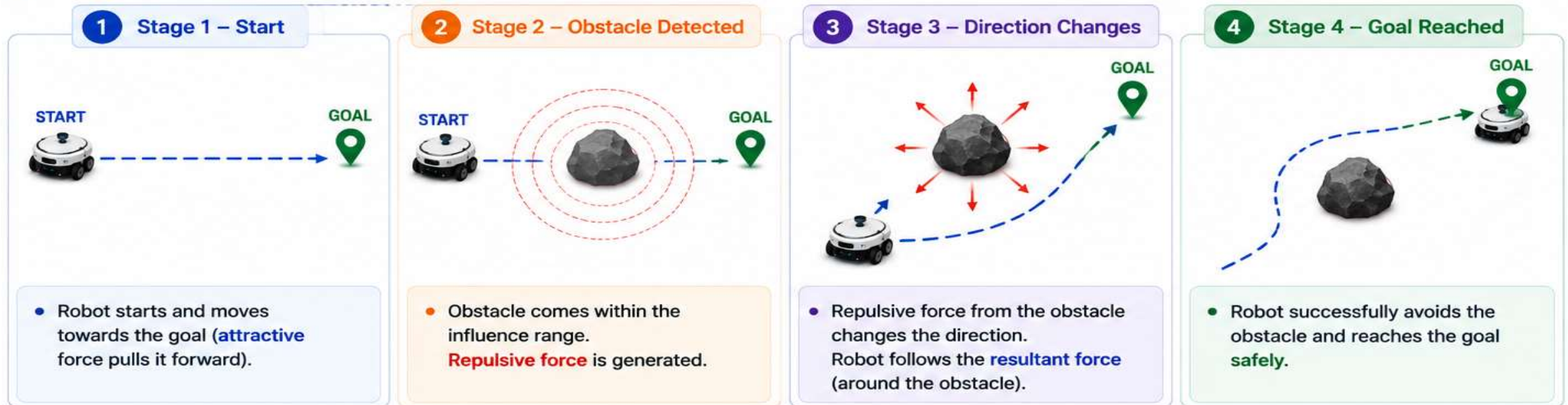


The APF algorithm works in a loop:

Sense → **Calculate** → **Move** → **Repeat**.

Can We Demonstrate APF with a Robot?

Example: A mobile robot navigates from the **start** position to the **goal** while **avoiding an obstacle** using Artificial Potential Field.



Key Message: The robot deviates from the direct path because the obstacle generates a **repulsive force**.

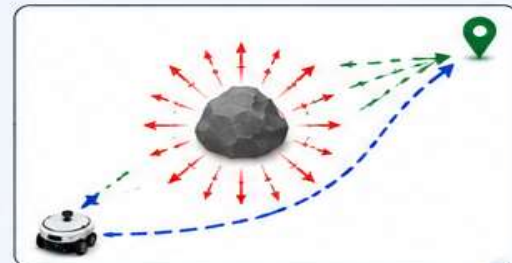
OPTIONAL: LIVE SIMULATION



A simple Python simulation can be used to visualize attractive (**green**), repulsive (**red**), and resultant (**blue**) forces in real time.

Typical Parameters

- Attractive gain (k_{att}) = 1.0
- Repulsive gain (k_{rep}) = 100.0
- Influence distance (d_0) = 2.0 m



Conclusion & Key Takeaways

The Artificial Potential Field (APF) method provides a **simple**, **efficient** and **effective** way for robots to navigate in dynamic environments.



KEY TAKEAWAYS



1. Attractive Force

The robot is pulled towards the goal using an **attractive potential field**.



2. Repulsive Force

Obstacles create **repulsive forces** that push the robot away when it comes too close.



3. Resultant Force

The **total force** (sum of attractive and repulsive forces) decides the robot's direction.



4. Continuous Process

The robot continuously **senses**, **calculates** and **moves** in a loop until the goal is reached.



5. Safe Navigation

APF enables smooth, real-time and **collision-free** navigation in unknown environments.



APPLICATIONS OF APF



• Mobile Robot Navigation

Autonomous robots in warehouses, hospitals, homes, and industries.



• UAV / Drone Path Planning

Safe navigation in cluttered or dynamic environments.



• Robot Manipulator Motion Planning

Obstacle avoidance in robot arms and pick-and-place tasks.



• Autonomous Vehicles

Local path planning and obstacle avoidance.



• Virtual & Simulation Environments

Used in games, VR, and training simulators.



ADVANTAGES

- ✓ Simple and easy to implement
- ✓ Real-time and computationally efficient
- ✓ Works well in static environments
- ✓ No need for global map (local information is enough)



LIMITATIONS

- ✗ Can get trapped in local minima
- ✗ Not suitable for highly dynamic environments
- ✗ Requires proper tuning of parameters



Bottom Line: APF is a powerful and intuitive technique that allows robots to reach their goal while avoiding obstacles by combining **attractive** and **repulsive** potential fields.



Next Steps & Future Scope

APF is a foundation for **intelligent navigation**. With enhancements, it can handle more **complex and real-world challenges**.

NEXT STEPS



- 1 Parameter Tuning**
Fine-tune attractive gain, repulsive gain and influence distance for **better performance**.



- 2 Dynamic Environments**
Extend APF to handle moving obstacles and changing goals in **real time**.



- 3 Hybrid Approaches**
Combine APF with other algorithms (e.g., A*, RRT, DWA) to overcome local minima and **improve global path planning**.



- 4 Advanced Sensing**
Integrate LiDAR, depth cameras and 3D sensors for **accurate environment perception**.



- 5 Real-World Deployment**
Implement and test on real robots in **real-world scenarios**.

FUTURE SCOPE OF APF



Useful in autonomous systems such as **self-driving cars, drones, and service robots**.



Can be applied in **smart cities, logistics, agriculture automation, and healthcare**.



Integration with **Artificial Intelligence and Machine Learning** for adaptive and predictive navigation.



Can be scaled for **multi-robot systems and swarm robotics**.

CHALLENGES TO ADDRESS



Local Minima Problem



Parameter Sensitivity



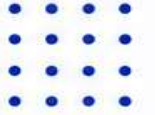
Dynamic Obstacles



Computational Efficiency



The Artificial Potential Field method is simple yet powerful. With continuous improvements, it can enable robots to navigate **smarter and safer**.





Thank You!

We hope you found our presentation
informative and insightful.



Any Questions?



GOAL



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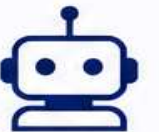
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Artificial Potential Field is a simple yet powerful approach that enables robots to navigate safely and efficiently towards their goal while avoiding obstacles.



**Smarter Paths.
Safer Robots.**



Thank you for your attention!

